NLTopt — companion notes: prescribed backbones, a density field, and supersonic panel flutter

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Status of these notes. These sections were split out of the main manuscript (`nltopt_paper.tex`) because they answer *different* questions from its thesis. They are preserved verbatim from the earlier draft. **Caveat:** every backbone result below was computed on the earlier *fixed-mode, 3-in-plane-mode* model, which the main paper shows under-resolves the in-plane relaxation of strongly steered fields (Sec. 5 there), and with an SSM master-eigenvector routine that has since been corrected for an ill-conditioning failure. Numbers here must be re-verified before any use. The flutter study (conforming BFS element, nonsymmetric pencil) is a separate paper in its own right: its question is a stability boundary, not a backbone, and its 9× figure is measured against a uniform layup chosen to sit in the bend–twist "flutter hole"; that baseline choice must be justified by a realistic frequency requirement before the figure can be claimed.

1 Dial-a-backbone: designing the toolpath to a target

Minimizing the detuning is one use of the adjoint; *prescribing* it is another, and it is the manufacturable analogue of the forced-response-curve design of [4, 3]. Replacing the objective with a least-squares match to a target relative detuning $\tau = J/J_{[0]}$ (the design's frequency-normalized hardening as a fraction of the unidirectional baseline's), the *same* tow-field adjoint steers a single panel to a *specified* nonlinearity. Figure 1 drives one panel to $\tau = 0.80, 0.65, 0.50$; the achieved values land on the targets to ~ 0.01, at held linear frequency and curl-feasibly, and the toolpath visibly steers harder for a deeper target (minimum tow radius $73 \to 30$ mm — the manufacturing cost of a stronger nonlinear correction). The panel becomes a *tunable* nonlinear element: a set of such panels, each dialed to spec, is a graded array of nonlinear resonators or matched absorbers.

2 Two fields: topology *and* fiber steering

The orientation field decides which way the tows run; the second wavetopo design field — the structural *density* $\boldsymbol{z}$ — decides where material goes at all. Adding it turns the problem into genuine topology optimization coupled to fiber steering, and it plugs into the same SSM backbone objective.

2.1 Three-field density and the density-weighted backbone

We use the standard density parametrization: a raw variable $z_e \in [0, 1]$ per element, a linear *filter* $\boldsymbol{y}_f = \boldsymbol{P}\boldsymbol{z}$ (radius R_ρ, imposing a minimum feature size), a smooth *Heaviside projection* $y = \frac{\tanh(\beta\eta)+\tanh(\beta(y_f-\eta))}{\tanh(\beta\eta)+\tanh(\beta(1-\eta))}$ that crisps the topology toward 0/1 as β is continued upward, and a

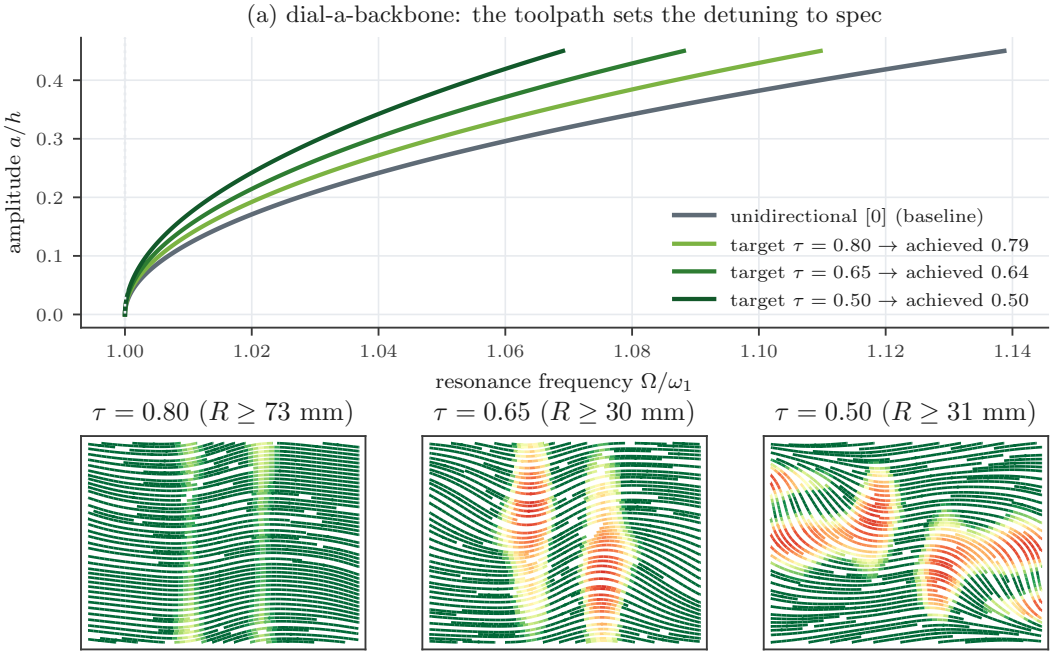


Figure 1: Designing the nonlinear response to a target. *Top:* one panel's first-mode backbone dialed to three prescribed relative detunings τ (a 20–50% reduction), each landing on target at held frequency. *Bottom:* the corresponding fiber toolpaths, coloured by local steering radius — a deeper target requires more aggressive (but still feasible) steering.

SIMP interpolation $E(y) = \varepsilon + (1-\varepsilon)y^p$ ($p = 3$) of the element stiffness. Every element quadratic form of Sec. **??** is scaled by E_e: the bending $K_b = \sum_e E_e\, k_e^b$, the raw cubic knl $= \sum_e E_e\, k_e^{nl}$, the in-plane stiffness $\boldsymbol{K}_{\text{in}} = \sum_e E_e\, \boldsymbol{K}_e^{\text{in}}$ and the coupling $\boldsymbol{g} = \sum_e E_e\, \boldsymbol{g}_e$, so the relaxed backbone coefficient $\gamma_{\text{eff}} = \text{knl} - 2\boldsymbol{g}^\top \boldsymbol{K}_{\text{in}}^{-1}\boldsymbol{g}$ (Eq. (**??**)) becomes a function of *both* fields. The modal mass $M = \rho h \sum_e y_e A_e \phi_e^2$ scales linearly with the physical density, so the linear frequency $\omega_1 = \sqrt{K_b/M}$ is available for the constraint below.

2.2 Coupled sensitivities

The design vector is $(\boldsymbol{z}, \hat{\boldsymbol{\theta}})$ and the objective is again $J = \gamma_{\text{eff}}/K_b$. The orientation sensitivity is Eq. (**??**) with each element term now carrying its weight E_e. The density sensitivity threads the three-field chain: with $\boldsymbol{u}^\star = \boldsymbol{K}_{\text{in}}^{-1}\boldsymbol{g}$,

$$\frac{\partial \gamma_{\text{eff}}}{\partial E_e} = k_e^{nl} - 4\,\boldsymbol{g}_e^\top \boldsymbol{u}^\star + 2\,\boldsymbol{u}^{\star\top} \boldsymbol{K}_e^{\text{in}} \boldsymbol{u}^\star, \qquad \frac{\mathrm{d}J}{\mathrm{d}\boldsymbol{z}} = \boldsymbol{P}^\top \Big[\frac{\partial J}{\partial E}\, \frac{\mathrm{d}E}{\mathrm{d}y}\, \frac{\mathrm{d}y}{\mathrm{d}y_f} \Big], \tag{1}$$

where $\partial J/\partial E = \partial_E \gamma_{\text{eff}}/K_b - \gamma_{\text{eff}} K_b^{-2} k_e^b$, $\mathrm{d}E/\mathrm{d}y = (1-\varepsilon)p\, y^{p-1}$ and $\mathrm{d}y/\mathrm{d}y_f$ is the Heaviside derivative — and the single adjoint solve $\boldsymbol{u}^\star$ is reused across every z_e and every $\hat{\theta}_j$. Both sensitivities are verified against central finite differences to $\lesssim 10^{-7}$; the orientation sensitivity also passes the *isotropy-annihilation* test — setting the lamina stiffness isotropic makes the tow angle inert and drives $\partial J/\partial \hat{\boldsymbol{\theta}}$ to $\sim 10^{-9}$, confirming the whole angular chain.

2.3 Constrained optimization

We drive $(\boldsymbol{z}, \hat{\boldsymbol{\theta}})$ with MMA under an augmented-Lagrangian for three constraints: the per-element tow-curvature limit $\zeta_e^2 \le \zeta_{\text{all}}^2$ (orientation only), a material-volume budget $\overline{y} = v_{\text{all}}$ as an *equality* (a fixed budget — without it the scale-near-invariant J would drive material trivially to zero), and a linear-frequency *band* $f_{\text{lo}}\,\omega_1^{\text{ref}} \le \omega_1 \le f_{\text{hi}}\,\omega_1^{\text{ref}}$. The band's *ceiling* is essential: with a floor alone the optimizer discovers the spurious low-density localized mode where the SIMP ε-stiffness with vanishing mass sends $\omega_1 \to \infty$ and $\gamma_{\text{eff}} \to 0$ trivially, giving a disconnected sliver design; the ceiling forbids it. Because the $(1,1)$ mode is symmetric about both centre-lines we project $\boldsymbol{z}$ onto the quarter-symmetric subspace, which selects the physical optimum over lopsided local minima.

2.4 Result

Figure 2 shows the converged two-field design at a held frequency and a fixed 50% material budget. The topology is connected and symmetric, and it voids material exactly at the top- and bottom-edge midpoints — the antinodes of the membrane stretch $(\nabla w)^2$, where the hardening integrand $\boldsymbol{\varepsilon}_{NL}^{\top}\boldsymbol{A}\boldsymbol{\varepsilon}_{NL}$ concentrates — while keeping the high-curvature centre that carries the frequency; the tows then steer *around* the voids, curl-feasibly. The first-mode backbone drift falls from 17.1% to 3.0% (-83%), extending the orientation-only lever.

Confirmation on the adapted mode. A single fixed master mode cannot reshape around the new voids, so one might expect it to over-credit topology. It does not: transferring the optimized topology onto the *real multi-mode* plate — SIMP-weighting its stiffness and mass at every quadrature point so the eigenmode itself adapts to the perforations — the SSM detuning reduction is 82%, against the reduced model's 83%. The relative-detuning lever therefore *survives* mode adaptation; it is not a fixed-mode artifact. One caveat we state plainly: the single-mode frequency is a loose proxy for the real multi-mode frequency of a strongly-steered design (the tow field alone reads far stiffer on the multi-mode plate, and the voids bring the frequency back to $0.94\times$ baseline), so the topology design's held-frequency rests on a cancellation rather than a tight constraint — constraining the real multi-mode frequency directly is the refinement. The contribution here is that the framework carries *both* manufacturable fields — orientation and topology — on the same verified SSM backbone objective and its adjoint, and that the topology lever holds up against the adapted-mode ground truth.

3 Designing the backbone to spec on the adapted mode

The one caveat left standing (Sec. 2) was that a *fixed* master mode mistracks the linear frequency of a strongly-steered design. We close it with a *mode-adaptive* reduction and use it to design the backbone to a target.

3.1 The adapted-mode reduction and its analytic gradient

Rather than freeze the master mode, take it to be the fundamental of the multi-mode Ritz eigenproblem $\boldsymbol{K}_b(\theta)\boldsymbol{v} = \omega_1^2\boldsymbol{M}\boldsymbol{v}$ over a small transverse basis $\{\phi_k\}$; the mode $w = \sum_k v_k\phi_k$ then reshapes with the design and ω_1 is the *real* multi-mode frequency. The relaxed cubic γ_{eff} (Eq. (**??**)) is evaluated on this adapted mode, and the objective is $J = \gamma_{\text{eff}}/(\omega_1^2\, m_c^2)$ with m_c the mode's midspan value (which fixes the physical amplitude normalization). The gradient is fully analytic and FD-verified to $\sim 10^{-7}$: the eigenvalue (frequency) sensitivity is Fox–Kapoor and needs no eigenvector

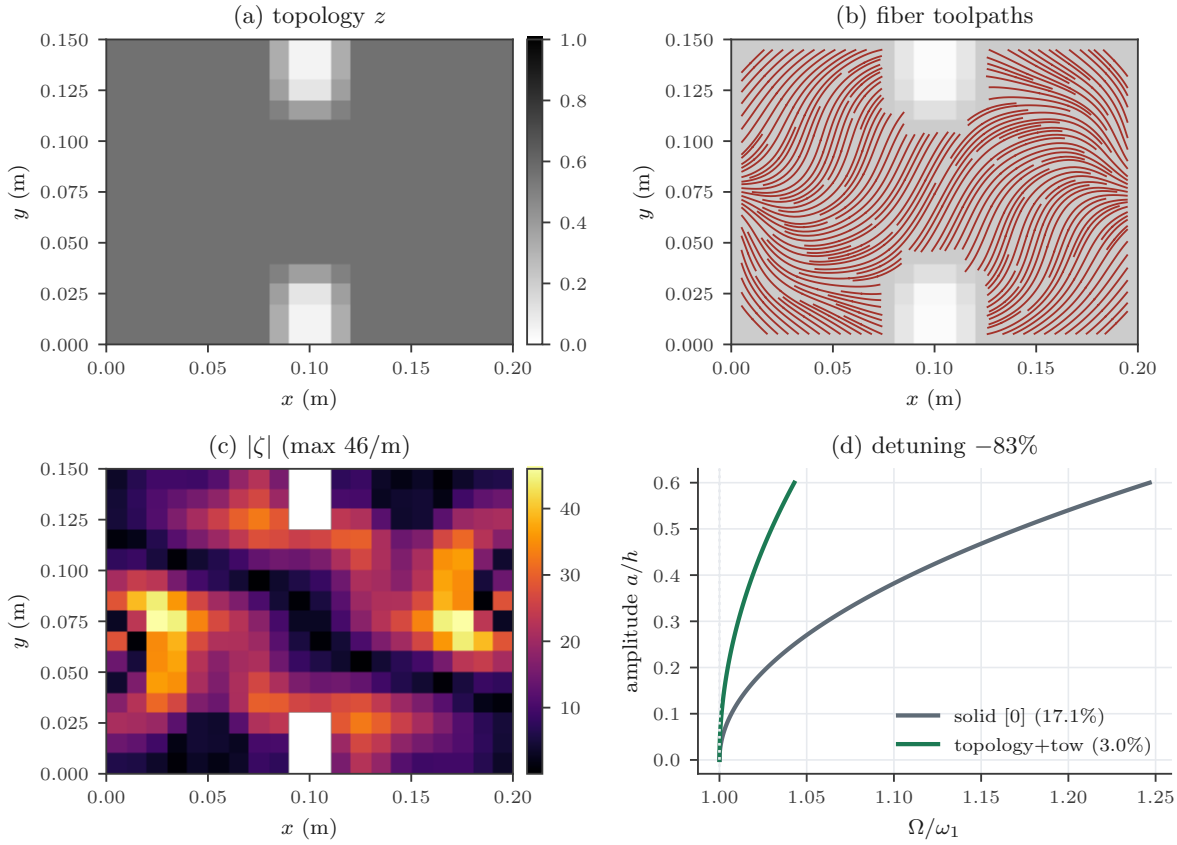


Figure 2: Two-field (topology + fiber-steering) design of the nonlinear backbone, rendered with wavetopo's tow/curl renderers. (a) the converged composite topology (density field $\boldsymbol{z}$); (b) the manufacturable fiber toolpaths, masked to the solid region and steering around the voids; (c) the tow curl $|\zeta|$ (curl-feasible, $R \geq 22\,\mathrm{mm}$); (d) the first-mode backbone of the solid unidirectional plate (grey) vs. the two-field design (green) at held frequency and fixed material budget — drift $17.1\% \to 3.0\%$. The voids sit at the membrane-stretch antinodes (the plate edges' midpoints), where the hardening integrand concentrates.

derivative, while the two eigenvector-sensitivity terms — of γ_{eff} and of m_c — each collapse to a *single adjoint curvature field* $\kappa_\beta = \sum_{j\neq 0} \beta_j \, \kappa_{v_j}$, so the whole sensitivity costs the one eigensolve plus cheap contractions.

3.2 Threading the backbone through design points

With both knobs in hand — the real ω_1 and the hardening curvature, each with a verified gradient — we can *place* the backbone: minimizing the squared mismatch to prescribed (f_k, a_k) points in the frequency–amplitude plane steers the tow field so the real backbone threads them. A caution the plate enforces: its cubic backbone is a two-parameter family (a linear frequency and one hardening curvature), so admissible design points must lie on *one* achievable, positively sloped hardening curve — a vertical (iso-frequency-across-amplitude) target is not a backbone a hardening plate can produce. Figure 3 designs an amplitude-stable resonator mandated near 335 Hz: the toolpath threads three points on a gentle-hardening target to an rms error of 3.3 Hz ($\sim$1%), curl-feasibly, far

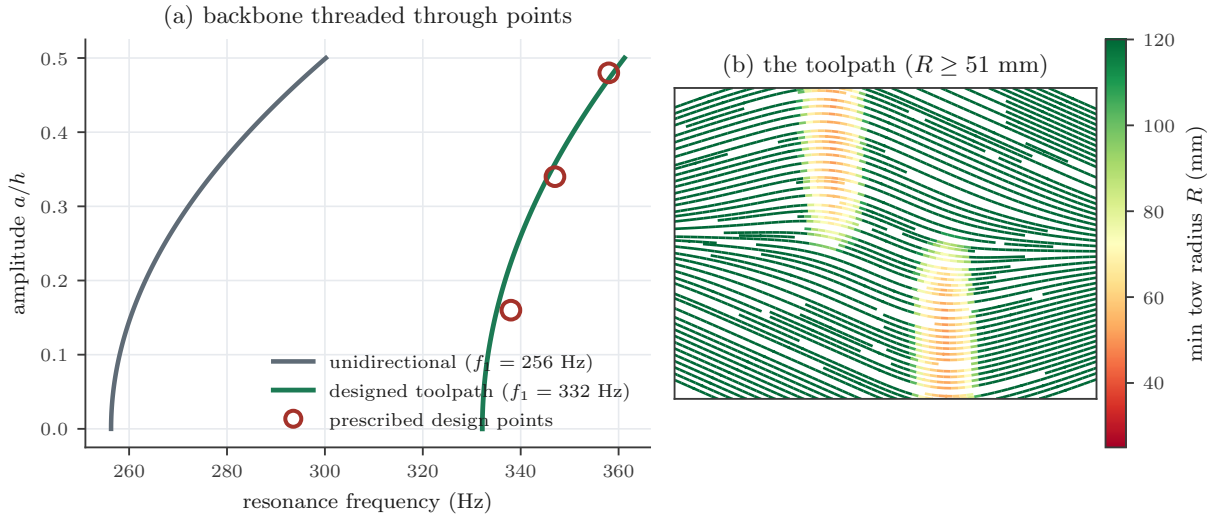


Figure 3: Designing a nonlinear backbone to spec. *Left:* the tow field is steered so the real (mode-adaptive) first-mode backbone threads three prescribed design points (circles) on a gentle-hardening target at a mandated $\sim$335 Hz — distinct from the unidirectional baseline (grey, 256 Hz, steep). *Right:* the manufacturable toolpath that does it, coloured by local steering radius. The reported frequency is the real multi-mode value (within 1.5% of the full plate), not a fixed-mode proxy.

flatter and higher than the unidirectional baseline. Crucially, the adapted-mode frequency (332 Hz) agrees with the full multi-mode plate (337 Hz) to 1.5% — where the old fixed-mode proxy erred by 70% — so the shipped device actually sits at its mandated frequency. This is the manufacturable, frequency-honest version of the response-curve design of [3, 4].

4 Designing an instability boundary: supersonic panel flutter

The backbone is one target the framework exposes; the *same* machinery designs a *stability boundary*. A thin panel in supersonic flow loses stability by flutter — a Hopf bifurcation of the flat state into self-excited oscillation — and keeping that boundary above the flight envelope is a hard constraint on every high-speed skin panel. Fiber steering is the natural manufacturable lever (this extends the variable-stiffness tailoring of [10, 11] from linear objectives to a nonlinear instability), and the geometric nonlinearity sets what happens *past* the boundary.

4.1 Aeroelastic model and calibration

First-order piston theory gives the aerodynamic pressure on a panel in a stream of Mach number M_∞ and dynamic pressure q; projected on the transverse modes it adds a *nonsymmetric* aerodynamic stiffness (the streamwise slope term) to the bending block,

$$G_{ij} = \int_\Omega \phi_i \, \partial_x \phi_j \, \mathrm{d}\Omega, \qquad \boldsymbol{M}\ddot{\boldsymbol{q}} + \boldsymbol{C}\dot{\boldsymbol{q}} + \big(\boldsymbol{K}_b(\theta) + \lambda_q \boldsymbol{G}\big)\boldsymbol{q} = \boldsymbol{0}, \quad \lambda_q = \frac{2q}{\sqrt{M_\infty^2 - 1}}, \tag{2}$$

plus light damping $\boldsymbol{C}$. Equation (2) is a nonsymmetric generalized eigenproblem; as q rises two modes coalesce in frequency and a complex-conjugate pair of the first-order pencil crosses the

imaginary axis at the critical q_{cr} — *coalescence flutter*. The implementation reproduces the classical result: a one-dimensional simply-supported panel gives the nondimensional critical dynamic pressure $\lambda_{\mathrm{cr}} = 343.35$ against the textbook 343.36 [8].

4.2 Conforming discretization: why the assumed modes fail here

Unlike the backbone, the flutter boundary is *not* adequately resolved by the global sine modes. A steered — or merely rotated — laminate develops bend–twist couplings D_{16}, D_{26} that *twist* the mode shapes; flutter is a coalescence of two modes and is exquisitely sensitive to their shape, and the sine basis, which cannot represent the twist, gives the wrong coalescence. We established this against an independent *conforming* Bogner–Fox–Schmit (BFS) bicubic-Hermite Kirchhoff element (nodal DOF $w, w_{,x}, w_{,y}, w_{,xy}$; frequencies exact against the closed form as in App. **??**): the two discretizations *agree* while $D_{16} \approx 0$ (0°–5°) and *diverge monotonically* as the coupling grows — by a factor of *ten* at 30°, where the sine model over-predicts q_{cr} tenfold. The flutter results below therefore use the conforming BFS element, and the sine-mode flutter is discarded. (The backbone results of Secs. **??**–3 are unaffected: those are frequencies and detuning *ratios*, FE-accurate to $\sim$2%, App. **??**.)

4.3 Steering the boundary out

On the conforming model the flutter boundary is a strong, *non-monotone* function of the ply angle: bend–twist coupling is maximally destabilizing near 30°–40°, where a uniform CFRP panel's q_{cr} collapses into a "flutter hole" an order of magnitude below the flow-aligned [0] value (Fig. 5a). A required operating frequency that places a uniform layup in that hole is exactly where steering helps: the linear frequency and the flutter boundary read the tow field through different operators, so a steered *field* can hold the frequency while placing the twist where it stabilizes rather than destabilizes. We maximize q_{cr} at held f_1 and mass, curl-feasibly, driving the search by the boundary's analytic adjoint — the implicit eigenvalue derivative

$$\frac{\mathrm{d}q_{\mathrm{cr}}}{\mathrm{d}\theta} = -\frac{\mathrm{Re}(\mathrm{d}s/\mathrm{d}\theta)}{\mathrm{Re}(\mathrm{d}s/\mathrm{d}q)}, \qquad \frac{\mathrm{d}s}{\mathrm{d}p} = \frac{\boldsymbol{u}^*(\partial_p \boldsymbol{\mathcal{A}})\,\boldsymbol{v}}{\boldsymbol{u}^* \boldsymbol{\mathcal{B}}\,\boldsymbol{v}}, \tag{3}$$

with $\boldsymbol{u}, \boldsymbol{v}$ the left/right eigenvectors of the critical mode of the (reduced) nonsymmetric aeroelastic pencil — the same left-eigenvector projection as the SSM adjoint (Eq. (**??**)), the reduction being built on a general pencil that never assumed self-adjointness. Formed on the conforming FE (reduced on its own modes, the modal-basis sensitivity absorbed by refreshing the basis each design step), it is FD-verified to 2.1×10^{-7}. On a $200 \times 150 \times 2$ mm CFRP panel at $M_\infty = 2$ and a held $f_1 = 300$ Hz — where the best uniform layup flutters at only $\sim$0.37 MPa — a curl-feasible steered field **recovers the flutter boundary to** 3.4 MPa**, a** 9× **margin** at held frequency and mass (Figs. 4b, 5b). Steering pushes the bending–torsion coalescence from $q_{\mathrm{cr}} = 0.37$ to 3.4 MPa (Fig. 4a), sliding the growth-rate zero-crossing — the Hopf onset — the same 9× to the right (Fig. 4b). The lever is the manufacturable placement of the same bend–twist coupling that, uncontrolled, craters the uniform panel.

4.4 The nonlinear aftermath

Past the boundary the geometric nonlinearity sets the consequence. Restoring the von Kármán relaxed cubic and reducing the flutter mode onto its Stuart–Landau normal form $\dot{p} = \lambda(q)\,p + \gamma\,p|p|^2$, the cubic coefficient — formed from the same left/right flutter eigenvectors and the von Kármán

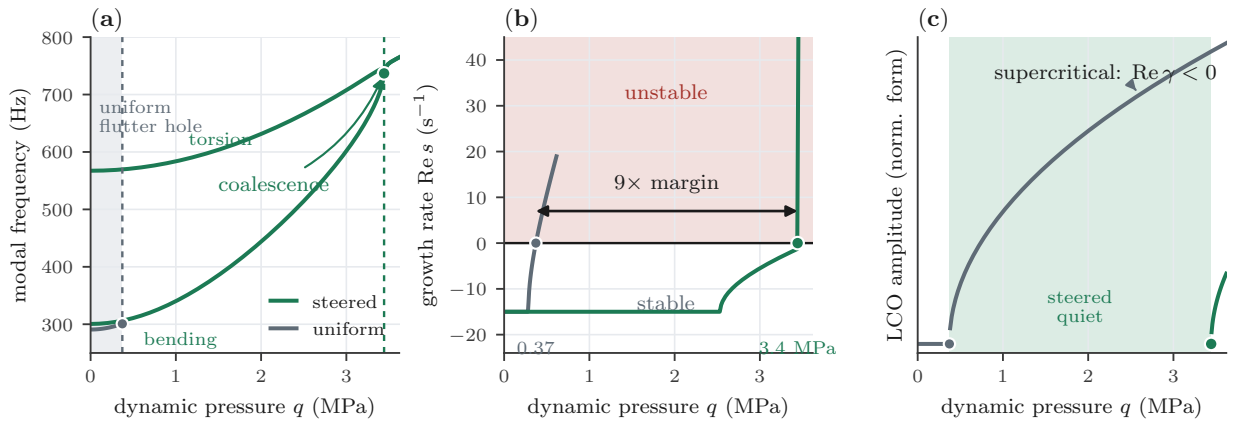


Figure 4: How steering reshapes the aeroelastic response of the panel (conforming BFS pencil; the best uniform layup at 300 Hz vs. the steered field). *(a)* modal frequency vs. dynamic pressure: the steered layup's bending and torsion branches coalesce far out in q (classic coalescence flutter, 745 Hz), whereas the uniform "flutter-hole" layup loses a single mode early. *(b)* growth rate $\mathrm{Re}\, s$ of the critical mode crosses zero — the Hopf onset — at q_{cr}; steering slides that crossing $\sim 9\times$ to higher q at held frequency and mass. *(c)* past the boundary $\mathrm{Re}\,\gamma < 0$ makes the Hopf supercritical, so the panel settles into a bounded limit cycle ($a \sim \sqrt{q - q_{\mathrm{cr}}}$, normal form); the shaded band is the recovered margin over which the steered panel stays quiet.

cubic tensor by the $\boldsymbol{u}_1^*$ projection — has $\mathrm{Re}\,\gamma < 0$: the Hopf is *supercritical*, so crossing it yields a benign, bounded limit-cycle oscillation rather than a catastrophic subcritical jump (Fig. 4c). This criticality — the sign of $\mathrm{Re}\,\gamma$ — is the cheap, model-robust output of the reduction. The finite limit-cycle amplitude on the conforming model, and its reduction by steering, are a direct follow-on (our earlier assumed-modes limit-cycle amplitudes are superseded along with the boundary they rested on).

4.5 Relation to the framework

Flutter uses no new reduction machinery — the nonsymmetric aerodynamic operator drops into the same first-order pencil, and the boundary sensitivity (3) is the same left-eigenvector projection as the backbone adjoint — but it did force a *conforming* discretization, and that rigor is the point: an independent FE cross-check overturned a plausible-looking sine-mode result and replaced a spurious $2\times$ with a correct, larger $9\times$ recovery. Linear flutter tailoring is itself classical aeroelasticity; the contribution here is its manufacturable, adjoint-driven form on a discretization honest enough to trust.

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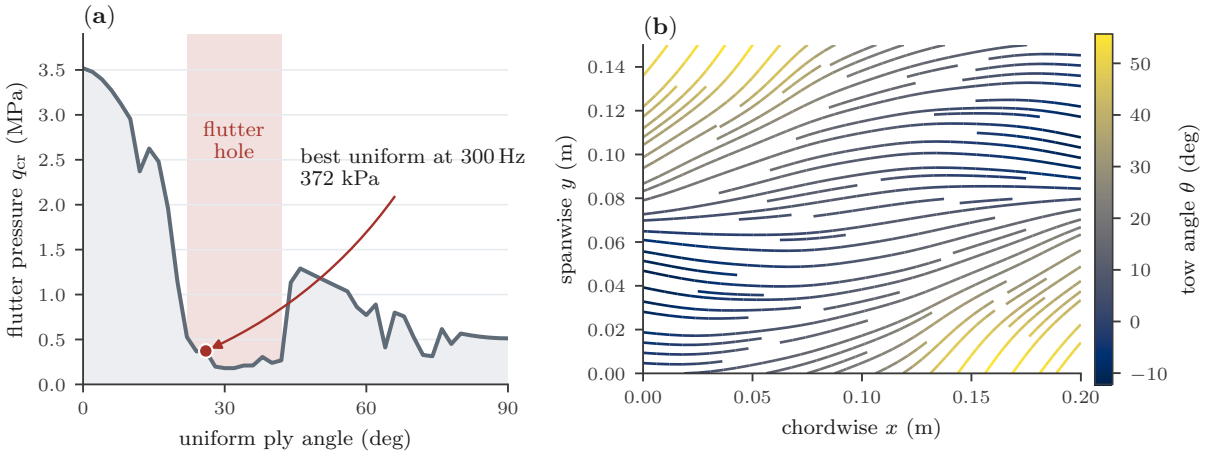


Figure 5: Flutter boundary on the conforming BFS FE. *(a)* the boundary vs. uniform ply angle collapses into a bend–twist "flutter hole" near 30°–40° (where D_{16} peaks), an order of magnitude below the flow-aligned value, so the best uniform layup that reaches the required 300 Hz flutters at only ∼0.37 MPa. *(b)* the manufacturable steered tow field that recovers the boundary to 3.4 MPa at held frequency and mass.

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